A 12-Year, 30 m Thermal Baseline for the Barcelona Metropolitan Area

A 12-year, 30 m thermal baseline for urban climate adaptation

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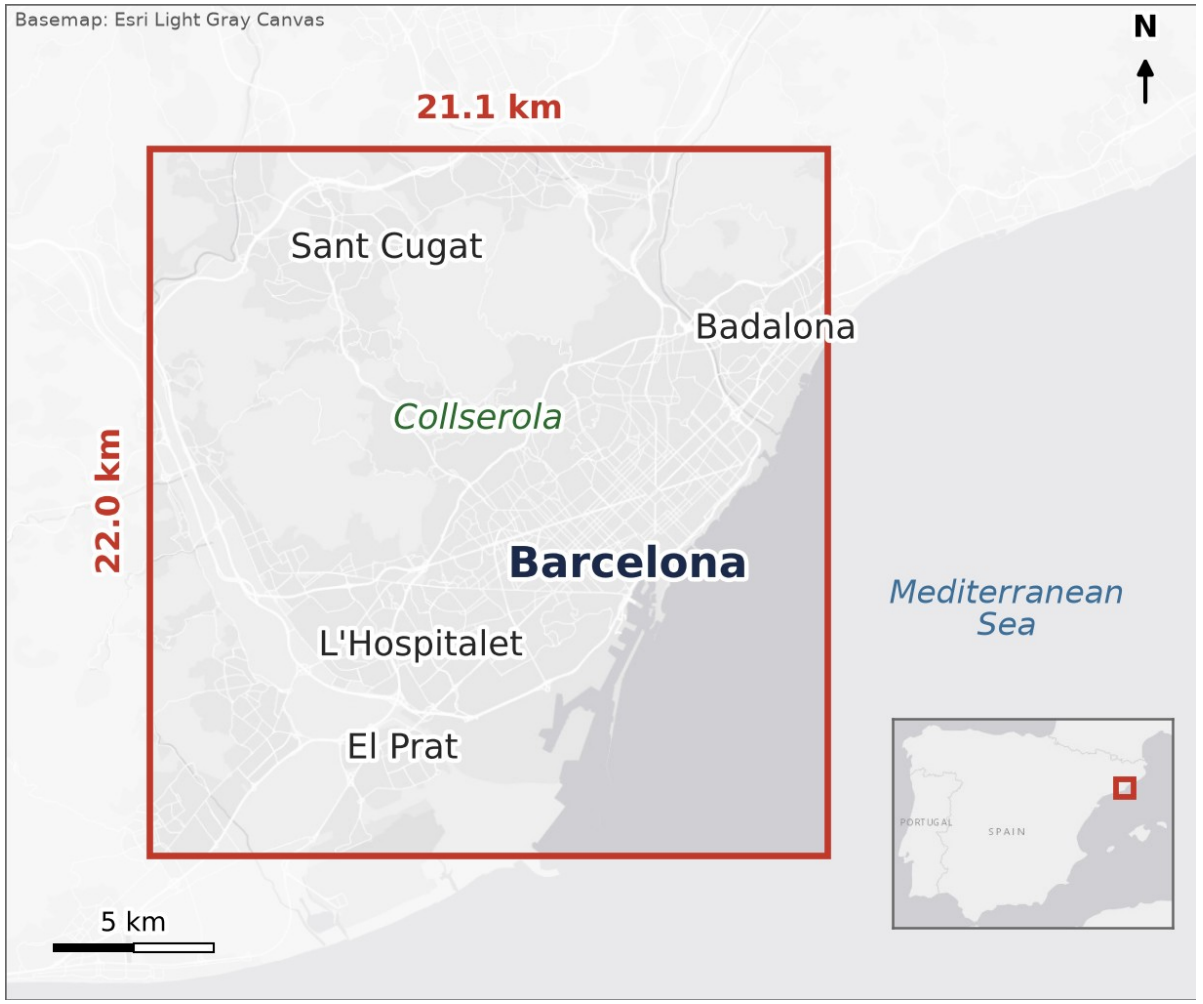
ECRS 2026 | Session S5 — Remote Sensing in Urban Areas | Oral, online

The Problem: No Baseline, No Way to Measure Change

- Outdoor thermal exposure depends strongly on urban surface temperature.
- The gap: long-term, high-resolution LST climatologies remain scarce for Mediterranean cities.
- Without a baseline: cannot measure change, cannot prioritize interventions, cannot tell which neighborhoods are persistently hotter.
- Barcelona: 3M+ people in its metropolitan area, a Mediterranean coastal city — until now, no systematic, satellite-based thermal baseline.

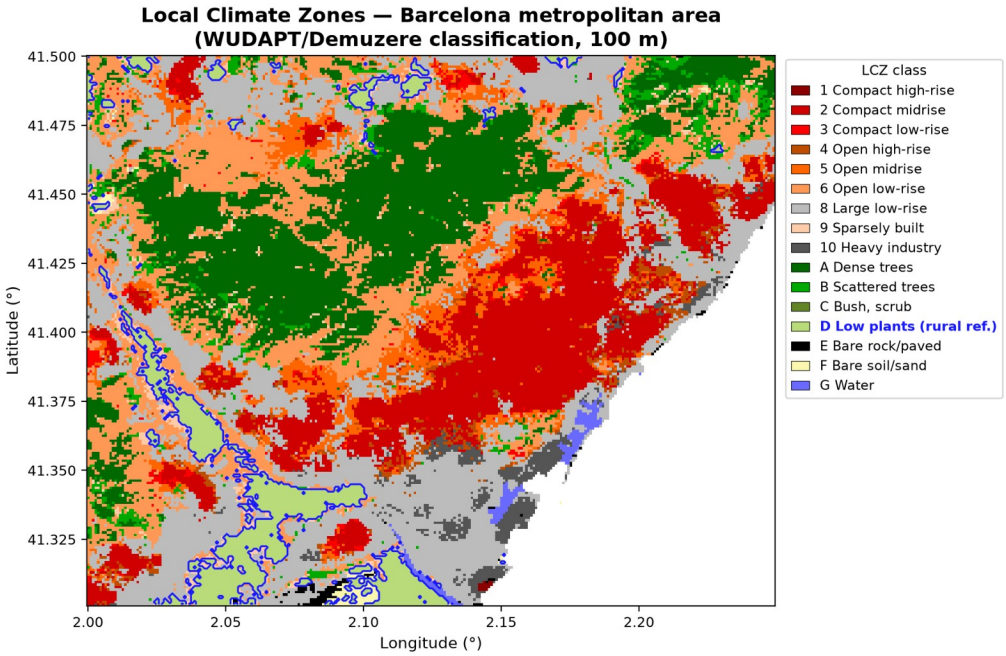
Study Area and Data

- 464 km² over the Barcelona metropolitan core (21.1 × 22.0 km).
- 378 Landsat 8/9 scenes, Collection 2 Level-2, 2013–2024 (12 years).
- 30 m grid, cloud-filtered.
- Four seasonal medians (winter, spring, summer, fall) → one climatology.



Local Climate Zones + Dual Consistency Checks

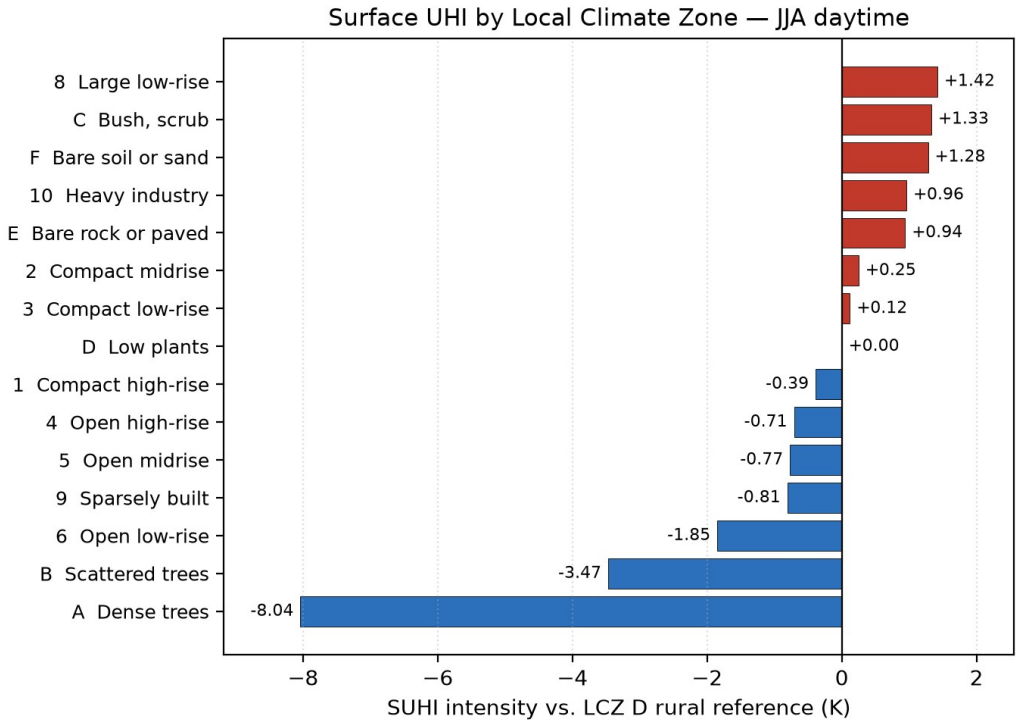
- Local Climate Zones (WUDAPT/Demuzere) — each zone has a distinct urban form: building height, density, vegetation cover.
- Reference: LCZ D (low plants) — the rural background.
- Two consistency checks, not one:
  - ERA5-Land skin temperature, n=325 matched pairs — checks the seasonal cycle.
  - ECOSTRESS, 70 m, n=7 near-coincident scene pairs (|Δt| ≤ 2h) — independent consistency assessment.
- Consistency — not validation. Seven scene pairs is not enough for full validation.



RESULTS

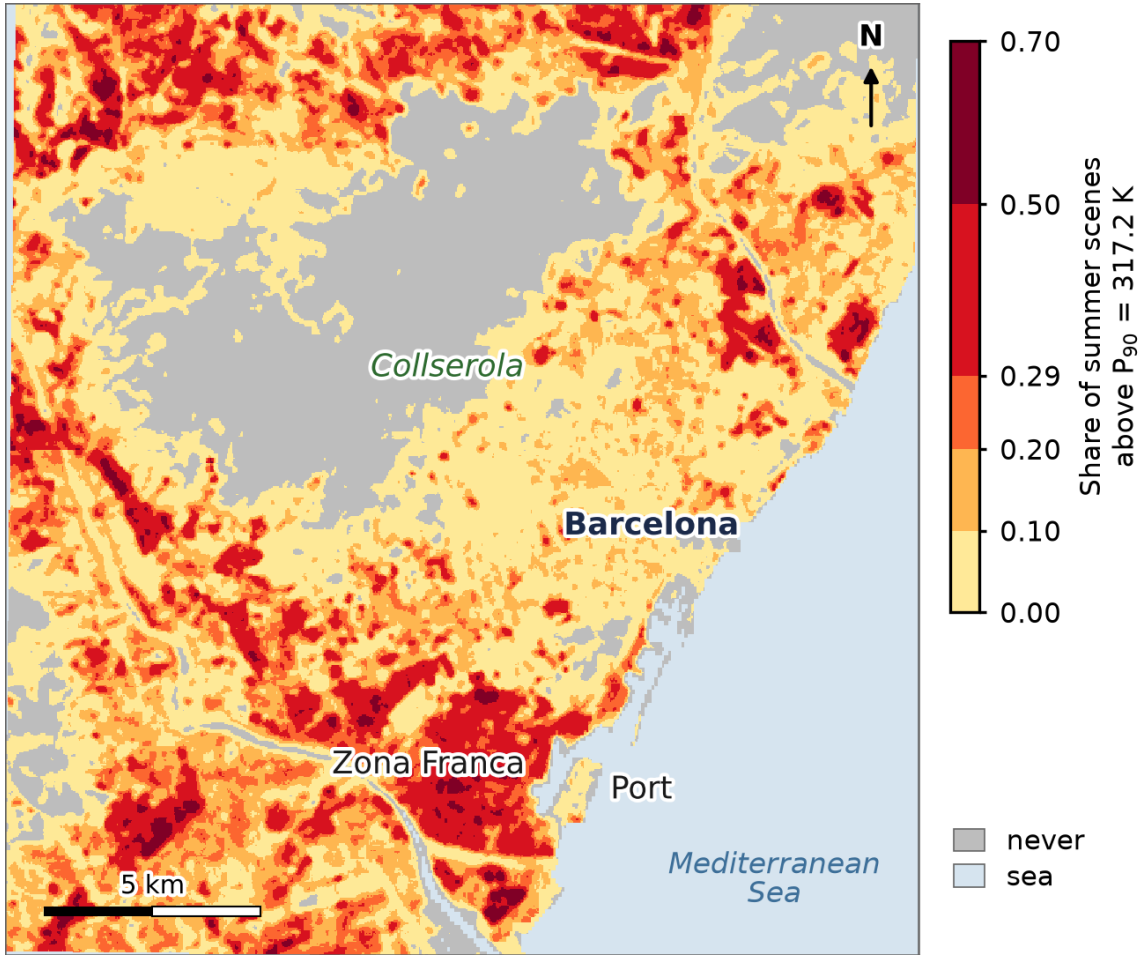
LCZ Thermal Hierarchy: The Tree-Canopy Cool Island

- Median JJA (summer) LST by Local Climate Zone, relative to LCZ D (rural reference):
  - LCZ 8 — large low-rise: +1.4 K (hottest built class)
  - LCZ 10 — heavy industry: +1.0 K
  - LCZ A — dense trees: −8.0 K (tree-canopy cool island)
- A 9.4-Kelvin gap between large low-rise and dense trees — same city, same 12-year summer climatology.
- It is not that built areas are exceptionally hot — it is that trees are dramatically cooler.



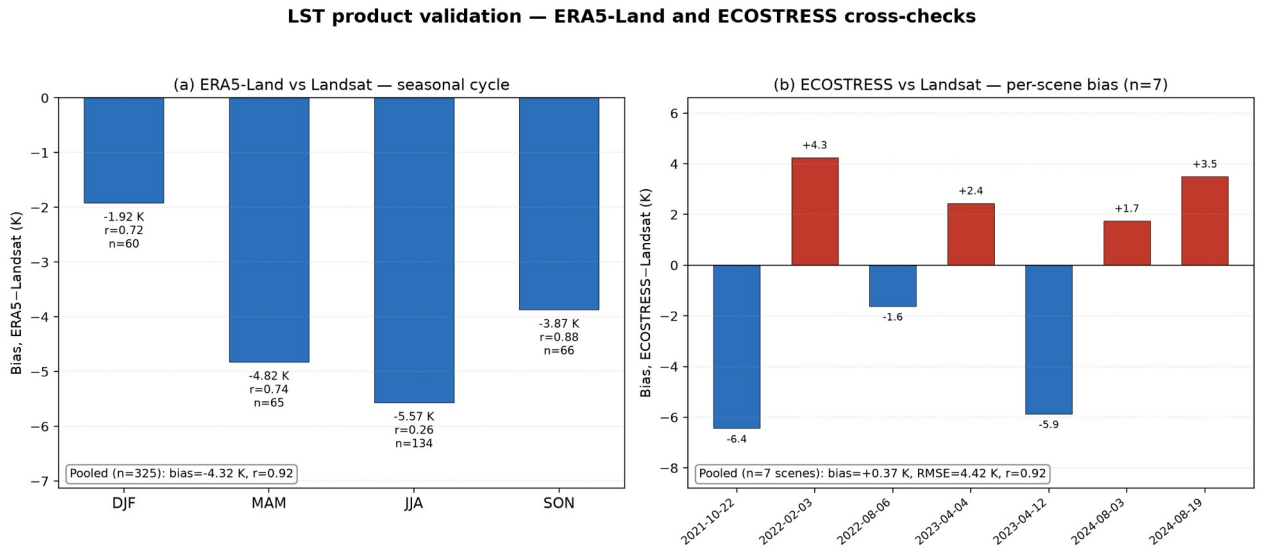
Hotspot Analysis: Where Heat Recurs

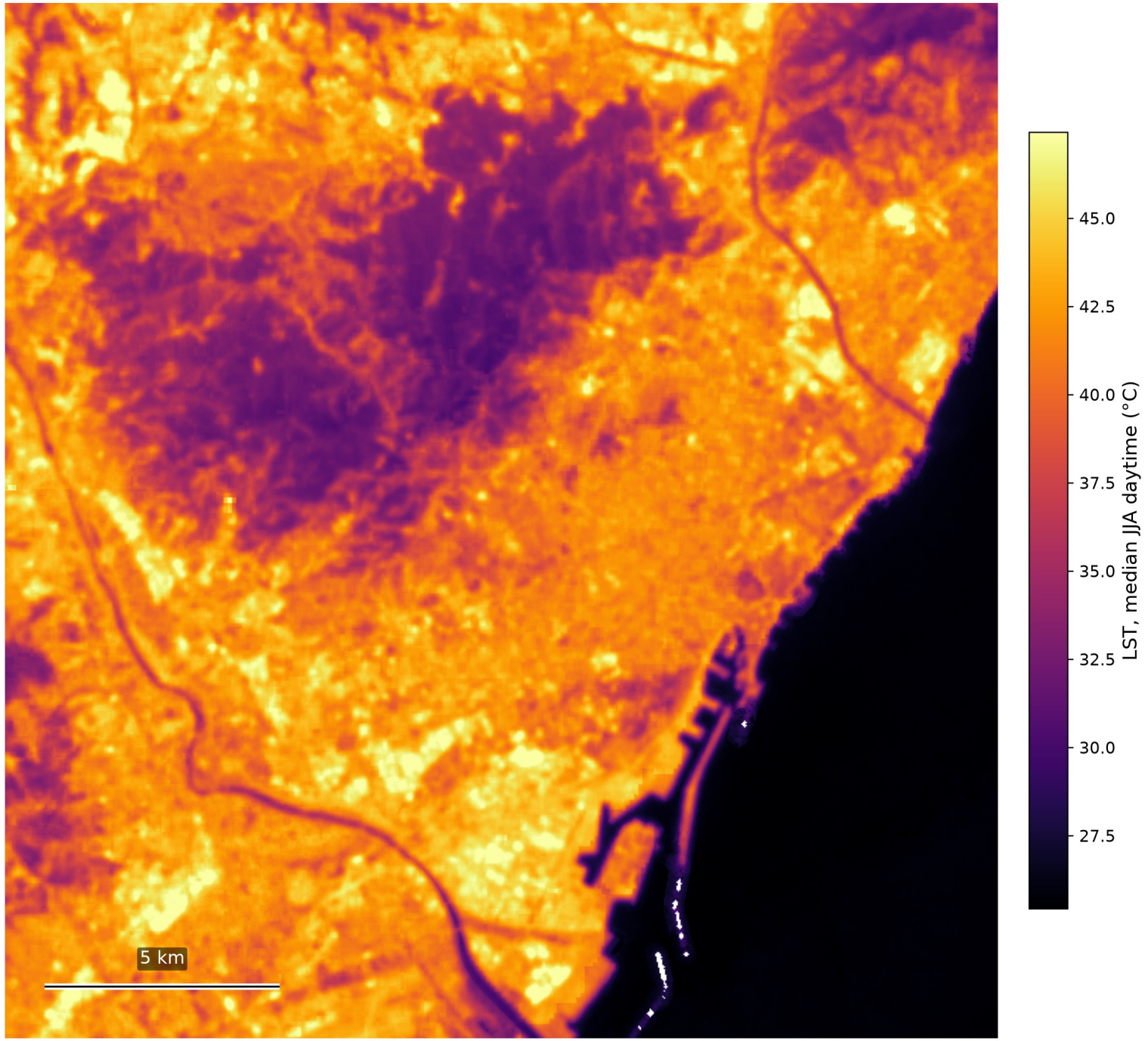
- Hotspot = pixel above the summer (JJA) P90 threshold.
- 78% of the land area (64% of the window, which includes sea) experiences at least one hotspot event in 12 years.
- 1.3% of the land area is persistently affected (hotspot frequency > 0.5) — the same pixels, year after year.
- These are the places where adaptation matters most: tree planting, cool roofs, urban redesign.



Two Independent Consistency Checks

- ERA5-Land (left panel): pooled seasonal cycle r = 0.92 — good agreement at the seasonal scale.
  - Within the summer season (JJA), r drops to 0.26 — ERA5 tracks the seasons but does not resolve intra-urban structure (0.1° grid).
- ECOSTRESS, 70 m (right panel): bias +0.37 K, RMSE 4.4 K, r = 0.92 — consistent with our 30 m field.
- ERA5 tells you what season it is. ECOSTRESS tells you your LST is about right. Only Landsat gives you the 30 m structure.



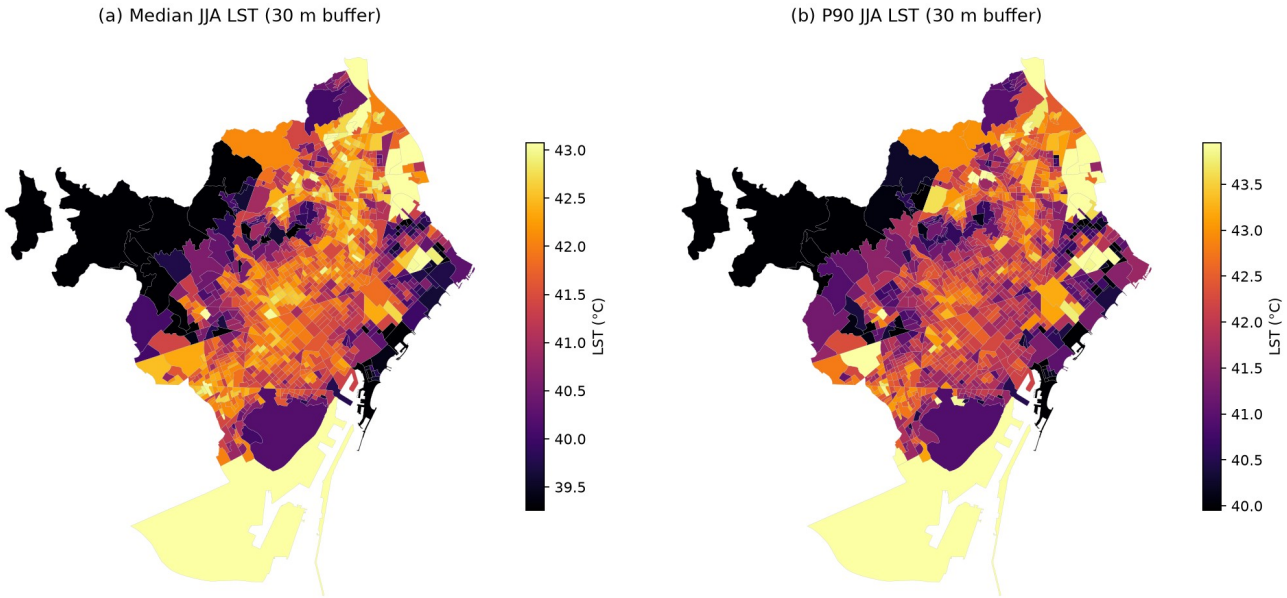


This is the baseline.

12 years. 30 meters.

Building Exposure: From Map to Baseline

- The climatology is not just a map — it is an exposure baseline.
- 106,000+ buildings overlaid across 13 municipalities; each building gets the median summer LST within a 30 m buffer.
- At Landsat's effective thermal resolution (~100 m), these are exposure indicators, not per-building retrievals.
- They tell you which neighborhoods carry the highest thermal load on their building stock.



Three Findings — and One Takeaway

- 1. A 12-year, 30 m SUHI-by-LCZ climatology for a Mediterranean city — reproducible, built entirely from open data.
- 2. Dense tree cover is 8 K cooler than the rural reference (≈9.4 K below large low-rise) — vegetation is the dominant spatial cooling signal.
- 3. ECOSTRESS confirms our Landsat field is consistent; ERA5 cannot substitute for high-resolution urban LST.

TAKEAWAY

A reproducible, satellite-based thermal baseline for adaptation planning in Mediterranean coastal cities.

Thank You

Slides, posters and figure code:

github.com/kerin84/ecrs2026-barcelona-lst

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Happy to take your questions.